Data Preparation

After exploring each feature, I found several missing values. For **ADDRTYPE** and **SEVERITY CODE column**, the rows containing missing values are dropped, because there are not many missing, value in these columns. For the other columns, each of them has an \Unknown" category. Hence, all the missing values in those columns are replaced with \Unknown" to avoid losing a lot of data.

Next, since all of the attributes are categorical, we transformed each of them into a binary-valued feature using one-hot encoding.

The target feature, SEVERITYCODE, contains several classes. They are class 0 for unknown severity, 1 for collisions that caused property damages, 2 for collisions with injuries, 2b for collisions with serious injuries, and 3 for collisions with fatalities. Because this project's goal is to prevent injuries or fatalities, I decided to re-group the data into two classes.

The first class, class 0, is for collisions with property damage (considered as not severe collisions). The second class, class 1, is for collisions with any injuries or even fatalities (severe collisions). I ignored the data with unknown severity code since those data will not help in predicting the severity of a collision. After re-grouping the data, I found out that the data distribution is imbalanced.

This will make the model biased towards the majority class. It is less important to accurately predict collisions with low severity, but it is more important to predict collisions with high severity. So, I apply the SMOTE (Synthetic Minority Oversampling Technique) to balance the data distribution.